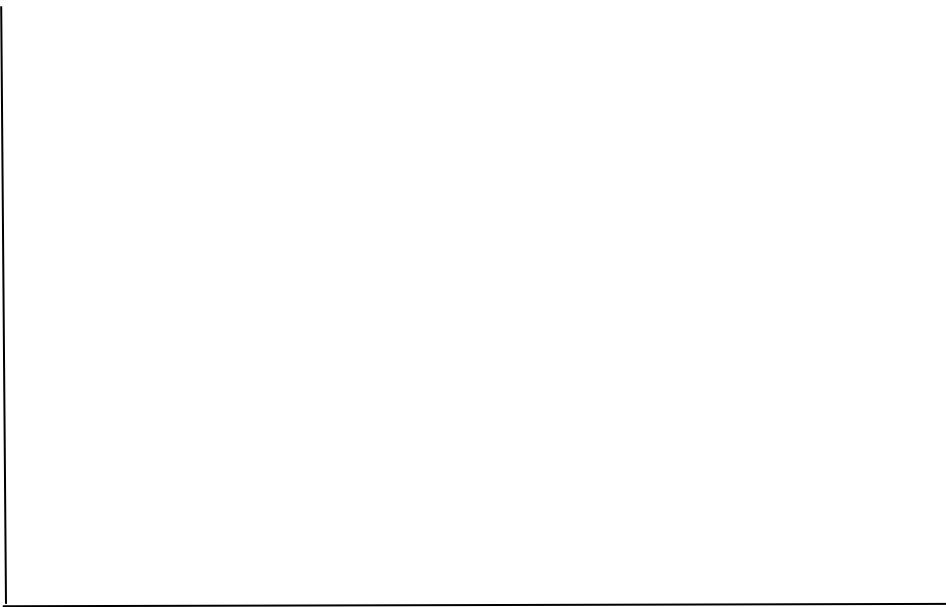


Unit 3 – Curve Sketching

Increasing & Decreasing Functions

- A function is **increasing** on an interval if the graph of the function **increases** from **left to right**.
- A function is **decreasing** on an interval if the graph of the function **decreases** from **left to right**.

Example: Locate the intervals of increase and decrease for the function below.



Increasing:

Decreasing:

- A function is **increasing** on an interval if the function has a **positive** slope of the tangent on that interval.
- A function is **decreasing** on an interval if the function has a **negative** slope of the tangent on that interval.

Test for Increasing and Decreasing Functions

- If $f'(x) > 0$ for all x in an interval, then $f(x)$ is **increasing** in that interval.
- If $f'(x) < 0$ for all x in an interval, then $f(x)$ is **decreasing** in that interval.

Critical Numbers

- A **critical number**, c , is a number in the domain of $f(x)$ such that

$$f'(c) = 0 \quad \text{OR} \quad f'(c) \text{ DNE.}$$

Example: Determine the intervals on which the following functions are increasing or decreasing.

a) $f(x) = 3x^2 + 6x - 1$

b) $f(x) = x^3 + 9x^2 + 24x - 3$

c) $f(x) = \frac{1}{4}x^4 + \frac{4}{3}x^3 - \frac{5}{2}x^2$